

(1)

Handouts - 1

1. given,

$$\begin{aligned}
 & \frac{1+2i}{3+4i} \\
 &= \frac{(1+2i)(3-4i)}{(3+4i)(3-4i)} \\
 &= \frac{3-4i+6i+8}{(3)^2-(4i)^2} \\
 &= \frac{11+2i}{25} = \frac{11}{25} + \frac{2}{25}i
 \end{aligned}$$

$$\text{Modulus, } |Z| = \sqrt{\left(\frac{11}{25}\right)^2 + \left(\frac{2}{25}\right)^2} = \frac{1}{\sqrt{5}}$$

$$\begin{aligned}
 \text{Argument, } \theta &= \tan^{-1} \left(\frac{\frac{2}{25}}{\frac{11}{25}} \right) = \tan^{-1} \left(\frac{2}{11} \right) \\
 &= 10.3^\circ
 \end{aligned}$$

$\therefore$ Thus, the modulus is $\frac{1}{\sqrt{5}}$ and argument is $\tan^{-1} \left(\frac{2}{11} \right)$.

3. given,

$$\begin{aligned}
 & \frac{6+i}{2-i} \\
 &= \frac{(6+i)(2+i)}{(2+i)(2-i)} \\
 &= \frac{12+6i+2i-1}{(2)^2-(i)^2} \\
 &= \frac{11+8i}{5} = \frac{11}{5} + \frac{8}{5}i
 \end{aligned}$$

$$\text{Modulus, } |z| = \sqrt{\left(\frac{11}{5}\right)^2 + \left(\frac{8}{5}\right)^2} = \frac{\sqrt{185}}{5}$$

$$\begin{aligned}\text{Argument, } \theta &= \tan^{-1} \left(\frac{\frac{8}{5}}{\frac{11}{5}} \right) \\ &= \tan^{-1} \left(\frac{8}{11} \right)\end{aligned}$$

$\therefore$ Thus, the modulus is $\frac{\sqrt{185}}{5}$ and argument is $\tan^{-1} \left(\frac{8}{11} \right)$.

5. given,

$$\begin{aligned}& \frac{3+4i}{2-3i} \\ &= \frac{(3+4i)(2+3i)}{(2+3i)(2-3i)} \\ &= \frac{6+9i+8i-12}{(2)^2 - (3i)^2} \\ &= \frac{-6+17i}{13} = -\frac{6}{13} + \frac{17}{13}i\end{aligned}$$

$$\begin{aligned}\text{modulus, } |z| &= \sqrt{\left(-\frac{6}{13}\right)^2 + \left(\frac{17}{13}\right)^2} \\ &= \frac{5}{\sqrt{13}}\end{aligned}$$

$$\begin{aligned}\text{argument, } \theta &= \pi - \tan^{-1} \left(\frac{\frac{17}{13}}{-\frac{6}{13}} \right) \\ &= \pi - \tan^{-1} \left(-\frac{17}{6} \right)\end{aligned}$$

$\therefore$ Thus, the modulus is $\frac{5}{\sqrt{13}}$ and the argument is $\pi - \tan^{-1} \left(-\frac{17}{6} \right)$.

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7. Given,

$$\begin{aligned} & \frac{2-i}{3+2i} \\ &= \frac{(2-i)(3-2i)}{(3+2i)(3-2i)} \\ &= \frac{6-4i-3i-2}{(3)^2-(2i)^2} \\ &= \frac{4-7i}{13} = \frac{4}{13} - \frac{7}{13}i \end{aligned}$$

$$\text{Modulus, } |Z| = \sqrt{\left(\frac{4}{13}\right)^2 + \left(-\frac{7}{13}\right)^2} = \frac{\sqrt{65}}{13} = \sqrt{\frac{5}{13}}$$

$$\begin{aligned} \text{Argument, } \theta &= -\tan^{-1}\left(\frac{-\frac{7}{13}}{\frac{4}{13}}\right) \\ &= -\tan^{-1}\left(\frac{7}{4}\right) \end{aligned}$$

$\therefore$ This modulus is $\sqrt{\frac{5}{13}}$ and argument $-\tan^{-1}\left(\frac{7}{4}\right)$.

9. Given,

$$\begin{aligned} \frac{5-2i}{2+5i} &= \frac{(5-2i)(2-5i)}{(2+5i)(2-5i)} \\ &= \frac{10-25i-4i-10}{(2)^2-(5i)^2} \\ &= \frac{8-49i}{29} = \frac{8}{29} - \frac{49}{29}i \end{aligned}$$

$$\text{Modulus, } |Z| = \sqrt{\left(\frac{8}{29}\right)^2 + \left(\frac{49}{29}\right)^2} = \sqrt{\frac{85}{29}}$$

$$\text{Argument, } \theta = -\tan^{-1}\left(\frac{\frac{49}{29}}{\frac{8}{29}}\right) = -\tan^{-1}\left(\frac{49}{8}\right)$$

$\therefore$ Thus, Modulus is $\sqrt{\frac{85}{29}}$ and argument is $-\tan^{-1}\left(\frac{49}{8}\right)$

1. given,

$$\begin{aligned} & \left(\frac{3+4i}{2-i} \right)^2 \\ &= \left(\frac{(3+4i)(2+i)}{(2+i)(2-i)} \right)^2 \\ &= \left\{ \frac{6+3i+8i-4}{(2)^2 - (i)^2} \right\}^2 \\ &= \left\{ \frac{2+11i}{5} \right\}^2 \\ &= \frac{(2+11i)^2}{25} \\ &= \frac{4+22i-121}{25} \\ &= \frac{-117+22i}{25} = -\frac{117}{25} + \frac{44}{25}i \end{aligned}$$

$$\begin{aligned} \text{Modulus, } |Z| &= \sqrt{\left(-\frac{117}{25}\right)^2 + \left(\frac{44}{25}\right)^2} \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{Argument, } \theta &= \pi - \tan^{-1} \left(\frac{\frac{44}{25}}{\frac{117}{25}} \right) \\ &= \pi - \tan^{-1} \left(\frac{44}{117} \right) \end{aligned}$$

Thus, the modulus is 5 and argument is $\pi - \tan^{-1} \left(\frac{44}{117} \right)$

3. given,

$$\frac{(5-2i)^3}{(2+i)^2}$$

$$= \frac{125 - 3 \cdot 5^2 \cdot 2i + 3 \cdot 5 \cdot (2i)^2 - (2i)^3}{4 + 4i + i^2}$$

$$= \frac{125 - 150i - 60 + 8i}{3 + 4i}$$

$$= \frac{(65 - 142i)(3 - 4i)}{(3 + 4i)(3 - 4i)}$$

$$= \frac{195 - 260i - 426i - 568}{3 - (4i)^2}$$

$$= \frac{-373 - 686i}{25} = -\frac{373}{25} - \frac{686}{25}i$$

$$\text{Modulus, } |Z| = \sqrt{\left(-\frac{373}{25}\right)^2 + \left(\frac{686}{25}\right)^2}$$

$$= \frac{29\sqrt{29}}{5}$$

$$\text{Argument, } \theta = -\pi + \tan^{-1}\left(\frac{\frac{686}{25}}{\frac{373}{25}}\right)$$

$$= -\pi + \tan^{-1}\left(\frac{686}{373}\right)$$

Thus, the modulus is $\frac{29\sqrt{29}}{5}$ and

argument is $-\pi + \tan^{-1}\left(\frac{686}{373}\right)$.

Cauchy-Riemann equation

1. given,

$$f(z) = z^3$$

$$= (x+iy)^3$$

$$= x^3 + 3ix^2y - 3xy^2 - iy^3$$

$$= (x^3 - 3xy^2) + i(3x^2y - y^3)$$

Equating real and imaginary parts,

$$u = x^3 - 3xy^2 \text{ and } v = 3x^2y - y^3$$

Now,

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2; \quad \frac{\partial u}{\partial y} = -6xy$$

$$\frac{\partial v}{\partial x} = 6xy; \quad \frac{\partial v}{\partial y} = 3x^2 - 3y^2$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$\therefore$ Cauchy-Riemann ^{equation} is satisfied. And

$f(z) = z^3$ is analytic

3. given,

$$f(z) = x^3 - 3xy^2 + i(3x^2y - y^3)$$

Equating real and imaginary parts,

$$u = x^3 - 3xy^2 \text{ and } v = 3x^2y - y^3$$

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2; \quad \frac{\partial u}{\partial y} = -6xy$$

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$$\frac{\partial v}{\partial x} = 6xy \quad ; \quad \frac{\partial v}{\partial y} = 3x^2 - 3y^2$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$\therefore$ Cauchy-Riemann equation is ~~not~~ satisfied.

5. given,

$$f(z) = x^2 + y^2 + i(2xy)$$

Equating real and imaginary parts,

$$u = x^2 + y^2 \quad \text{and} \quad v = 2xy$$

$$\text{now, } \frac{\partial u}{\partial x} = 2x \quad ; \quad \frac{\partial u}{\partial y} = 2y$$

$$\frac{\partial v}{\partial x} = 2y \quad ; \quad \frac{\partial v}{\partial y} = 2x$$

Here, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ but $\frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x}$; Thus, Cauchy-Riemann equation doesn't satisfy. So, $f(z)$ is not analytic.

7. given,

$$f(z) = \frac{x-iy}{x+iy}$$

$$= \frac{(x-iy)(x-iy)}{(x+iy)(x-iy)}$$

$$= \frac{(x-iy)^2}{x^2 - (iy)^2}$$

$$= \frac{x^2 - 2ixy - y^2}{x^2 + y^2}$$

$$= \frac{x^2 - y^2}{x^2 + y^2} + i \frac{-2xy}{x^2 + y^2}$$

Equating real and imaginary parts,

$$u = \frac{x^2 - y^2}{x^2 + y^2} \text{ and } v = \frac{-2xy}{x^2 + y^2}$$

$$\frac{\partial u}{\partial x} = \frac{(x^2 + y^2) 2x - (x^2 - y^2) 2x}{(x^2 + y^2)^2}$$

$$= \frac{4xy^2}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{(x^2 + y^2)(-2y) - (x^2 - y^2)(2y)}{(x^2 + y^2)^2}$$

$$= \frac{-4x^2y}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial x} = \frac{(x^2 + y^2)(-2y) - (-2xy) 2x}{(x^2 + y^2)^2}$$

$$= \frac{2x^2y - 2y^3}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{-2x(x^2 + y^2) - (-2xy) 2y}{(x^2 + y^2)^2}$$

$$= \frac{2xy^2 - 2x^3}{(x^2 + y^2)^2}$$

Here, $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} \neq -\frac{\partial v}{\partial x}$; Thus, Cauchy-Riemann equations doesn't satisfy.

9. given,

$$f(z) = e^{-y}(\cos x + i \sin x)$$
$$= e^{-y} \cos x + i e^{-y} \sin x$$

Equating real and imaginary parts,

$$u = e^{-y} \cos x \text{ and } v = e^{-y} \sin x$$

now,

$$\frac{\partial u}{\partial x} = -e^{-y} \sin x ; \frac{\partial u}{\partial y} = -e^{-y} \cos x$$

$$\frac{\partial v}{\partial x} = e^{-y} \cos x ; \frac{\partial v}{\partial y} = -e^{-y} \sin x$$

$$\text{Here, } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Thus, Cauchy-Riemann equations satisfy.

Harmonic function and Their Conjugate

problem-1:

given,

$$u = x^2 - y^2 + 4x - 3y$$

$$\therefore \frac{\partial u}{\partial x} = 2x + 4 \dots \textcircled{1}$$

$$\frac{\partial u}{\partial y} = -2y - 3 \dots \textcircled{2}$$

$$\frac{\partial^2 u}{\partial x^2} = 2 \dots \textcircled{3}$$

$$\frac{\partial^2 u}{\partial y^2} = -2 \dots \textcircled{4}$$

$$\textcircled{3} + \textcircled{4} \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$\therefore u$ is harmonic.

For finding v :

By Cauchy - Riemann eqn, we have

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 2x + 4 \dots \textcircled{5}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -2y - 3 \dots \textcircled{6}$$

$$\begin{aligned} \textcircled{5} \Rightarrow v &= \int (2x + 4) dy & \left[\begin{array}{l} \text{Integration w.r.t. } y \\ \text{treating } x \text{ as constant} \end{array} \right] \\ &= 2xy + 4y + F(x) \dots \textcircled{7} \end{aligned}$$

$$\Rightarrow \frac{\partial v}{\partial x} = 2y + F'(x)$$

$$\Rightarrow F'(x) = \cancel{2y} + 3 - 2y$$

$$\Rightarrow F'(x) = 3$$

$$\Rightarrow F(x) = \int 3 dx = 3x + C$$

$$\textcircled{7} \Rightarrow v = 2xy + 4y + 3x + C$$

(Ans.)

problem 3: given,

$$u = \ln(x^2 + y^2)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2} \quad \dots (1)$$

$$\frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2} \quad \dots (2)$$

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{(x^2 + y^2) \cdot 2 - 2x(2x)}{(x^2 + y^2)^2} \\ &= \frac{-2x^2 + 2y^2}{(x^2 + y^2)^2} \quad \dots (3) \end{aligned}$$

$$\begin{aligned} \frac{\partial^2 u}{\partial y^2} &= \frac{(x^2 + y^2) \cdot 2 - 2y(2y)}{(x^2 + y^2)^2} \\ &= \frac{2x^2 - 2y^2}{(x^2 + y^2)^2} \quad \dots (4) \end{aligned}$$

$$\text{Now, } (3) + (4) \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$\therefore u$ is harmonic.

for finding v ,

By Cauchy-Riemann eqn, we get,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{2x}{x^2 + y^2} \quad \dots (5)$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = \frac{2y}{x^2 + y^2} \quad \dots (6)$$

$$(5) \Rightarrow V = \int \frac{2x}{x^2 + y^2} dy$$

[Integration w.r.t. y treating x as constant]

$$= 2x \cdot \frac{1}{x} \cdot \tan^{-1}\left(\frac{y}{x}\right) + F(x)$$

$$= 2 \tan^{-1}\left(\frac{y}{x}\right) + F(x) \dots (6)$$

$$\Rightarrow \frac{\partial V}{\partial x} = 2 \cdot \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \left(-\frac{y}{x^2}\right) + F'(x)$$

$$\Rightarrow -\frac{2y}{x^2 + y^2} = -\frac{2y}{x^2 + y^2} + F'(x)$$

$$\Rightarrow F'(x) = 0$$

$$\therefore F(x) = C$$

$$(7) \Rightarrow V(x, y) = 2 \tan^{-1}\left(\frac{y}{x}\right) + C \quad (\text{Ans})$$

Prob lem 5: given,

$$u = x^3 - 3xy^2$$

$$\therefore \frac{\partial u}{\partial x} = 3x^2 - 3y^2 \dots (1)$$

$$\frac{\partial u}{\partial y} = -36xy \dots (2)$$

$$\frac{\partial^2 u}{\partial x^2} = 6x \dots (3)$$

$$\frac{\partial^2 u}{\partial y^2} = -6x \dots (4)$$

Here, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

$\therefore u$ is harmonic.

for finding v ,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 3x^2 - 3y^2 \dots (5)$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -6xy \dots (6)$$

$$(5) \Rightarrow v = \int (3x^2 - 3y^2) dy$$

[Integration w.r.t. y
treating x as constant]

$$= 3xy - y^3 + F(x) \dots (7)$$

$$\Rightarrow \frac{\partial v}{\partial x} = 6xy + F'(x)$$

$$\Rightarrow 6xy = 6xy + F'(x)$$

$$\Rightarrow F'(x) = 0$$

$$\Rightarrow F(x) = C$$

$$(7) \Rightarrow v(x, y) = 3xy - y^3 + C$$

(Ans)

Problem 7: given,

$$u = \frac{x}{x^2 + y^2}$$

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{(x^2 + y^2) - x(2x)}{(x^2 + y^2)^2} \\ &= \frac{-x^2 + y^2}{(x^2 + y^2)^2} \quad \text{--- (1)} \end{aligned}$$

$$\begin{aligned} \frac{\partial u}{\partial y} &= \frac{(x^2 + y^2) \cdot 0 - x(2y)}{(x^2 + y^2)^2} \\ &= \frac{-2xy}{(x^2 + y^2)^2} \quad \text{--- (2)} \end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{(x^2 + y^2)^2(-2x) - (-x^2 + y^2)2 \cdot (x^2 + y^2) \cdot 2x}{(x^2 + y^2)^4}$$

$$= \frac{-5x^5 - 10x^3y^2 - 5y^5 +}{(x^2 + y^2)^4}$$

$$= \frac{-4x^5 - 8x^3y^2 - 4y^5 + 4x^5 + 4x^3y^2 - 4}{(x^2 + y^2)^4}$$

$$= \frac{-2x(x^2 + y^2) - 4x(-x^2 + y^2)}{(x^2 + y^2)^3}$$

$$= \frac{-2x^3 - 2xy^2 + 4x^3 - 4xy^2}{(x^2 + y^2)^3}$$

$$= \frac{2x^3 - 6xy^2}{(x^2 + y^2)^3} \quad \text{--- (3)}$$

$$\frac{\partial^4 u}{\partial y^4} = \frac{(x^2+y^2)^2(-2x) - (-2xy) \cdot 2(x^2+y^2)(2y)}{(x^2+y^2)^4}$$

$$= \frac{-2x(x^2+y^2) + 8xy^2}{(x^2+y^2)^3}$$

$$= \frac{-2x^3 - 2xy^2 + 6xy^2}{(x^2+y^2)^3}$$

$$= \frac{-2x^3 + 6xy^2}{(x^2+y^2)^3}$$

$$\text{Here, } \frac{\partial^4 u}{\partial x^4} + \frac{\partial^4 u}{\partial y^4} = 0$$

$\therefore u$ is harmonic.

For finding v , using Cauchy-Riemann equation.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{-x^2 + y^2}{(x^2 + y^2)^2} \quad \dots \quad (5)$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = \frac{-2xy}{(x^2 + y^2)^2} \quad \dots \quad (6)$$

Integration Eq. (5) w.r.t. y and treating x as constant,

$$v = \int \frac{-x^2 + y^2}{(x^2 + y^2)^2} dy$$

$$\Rightarrow v = -\frac{xy}{x^2 + y^2} + F(x) \quad \dots \quad (7)$$

$$\frac{\partial V}{\partial x} = \frac{2xy}{(x^2+y^2)^2} + g'f'(x)$$

$$\Rightarrow \frac{2xy}{(x^2+y^2)^2} = \frac{2xy}{(x^2+y^2)^2} + f'(x)$$

$$\Rightarrow f'(x) = 0$$

$$\Rightarrow f(x) = C$$

$$\therefore V(x, y) = -\frac{y}{(x^2+y^2)} + C \quad (\text{Ans}),$$

Cauchy's Integral formula

1. given, $\oint_C \frac{e^z}{z-2} dz$

where C is the circle $|z|=3$

(i) C is the circle with radius of 3 and centered at the origin and $|a|=2$ which lies inside of the circle.

(ii) The function $f(z) = e^z$ is which is analytic everywhere.

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Applying Cauchy's Integral formula:

$$\oint_C \frac{e^z}{z-2} dz = 2\pi i f(2)$$

$$= 2\pi i \times e^2$$

$$= 2\pi i e^2 \quad (\text{Ans.})$$

3. given, $\oint_C \frac{e^{iz}}{z^4} dz$

where C is circle $|z|=1$

(i) Here closed curve C is a circle with radius 1 and centered at the origin and $a=0$ which lies inside the circle.

(ii) The function $f(z) = e^{iz}$ is an exponential function which is analytic everywhere.

Applying Cauchy's generalized integral formula:

$$\oint_C \frac{e^{iz}}{z^n} dz = \frac{2\pi i}{(n-1)!} \frac{d^{n-1}}{dz^{n-1}} (e^{iz}) \Big|_{z=0} \quad \left| \begin{array}{l} \text{Here} \\ n=1 \end{array} \right.$$

$$= 2\pi i [ie^{iz}]_{z=0}$$

$$= 2\pi i (ie^{i \cdot 0})$$

$$= 2\pi i \cdot i$$

$$= -2\pi \text{ (Ans.)}$$

5. Given, $\oint_C \frac{\sin z}{(z - \frac{\pi}{6})^4} dz$

where C is circle $|z| = 2$.

(i) The closed curve C is a circle with radius 2 centered at the origin and $a = \frac{\pi}{6}$ which lies inside the circle.

(ii) The function $f(z) = \sin z$ is a trigonometric function which is analytic everywhere.

Computing derivatives of $f(z)$:

$$f'(z) = \cos z$$

$$f''(z) = -\sin z$$

$$f'''(z) = -\cos z$$

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Applying Cauchy's generalized integral formula:

$$\oint_C \frac{\sin z}{(z - \frac{\pi}{6})^4} = \frac{2\pi i}{3!} \frac{d^3}{dz^3} [f(z)]_{z=a=\frac{\pi}{6}} \quad \left| \begin{array}{l} \text{Here,} \\ n=3 \end{array} \right.$$

$$= \frac{2\pi i}{6} [f'''(z)]_{z=\frac{\pi}{6}}$$

$$= \frac{2\pi i}{6} [-\cos z]_{z=\frac{\pi}{6}}$$

$$= \frac{2\pi i}{6} [-\cos(\frac{\pi}{6})]$$

$$= -\frac{\sqrt{3}\pi i}{6} \quad (\text{Ans.})$$

7. given, $\oint_C \frac{\log(1+z)}{z^2} dz$

where C is the circle $|z|=1$

(i) The closed curve C is the circle with radius 1 and centered at the origin and $a=0$ which lies inside the closed curve.

(ii) The function $f(z) = \log(1+z)$ is analytic inside and on the closed curve when $z > -1$.

Applying Cauchy's generalized integral formula!

$$\oint_C \frac{\log(1+z)}{z^2} dz = \frac{2\pi i}{1!} \frac{d}{dz} [\log(1+z)]_{z=0} \quad \text{Here, } n=1$$

$$= 2\pi i \left[\frac{1}{1+z} \right]_{z=0}$$

$$= 2\pi i \quad \text{Ans.}$$

9. Given, $\oint_C \frac{\cos z}{z^2 - 4} dz$

where C is the square $x = \pm 2$ and $y = \pm 2$.

Here, the square is $x = \pm 2$ and $y = \pm 2$ and the maximum distance from origin is $2\sqrt{2}$; which is $a = 4$ so,

$$2\sqrt{2} < 4.$$

$\therefore$ So, by Cauchy's theorem;

$$\oint_C \frac{\cos z}{z^2 - 4} dz = 0$$

(Ans.)

Handouts 2
Complex variable

Problem-18 given,

$$f(z) = \frac{z^3 + 2z}{(z-2)(z^2+1)} = \frac{z^3 + 2z}{(z-2)(z+i)(z-i)}$$

when, denominator is zero,

$$(z-2)(z^2+1) = 0$$

Then, $f(z)$ has singularity at $z=2, +i, -i$.

Hence, when $z=2$ at ^{pole of} order 1.

$$\lim_{z \rightarrow 2} (z-2) \frac{z^3 + 2z}{(z-2)(z^2+1)}$$

$$= \lim_{z \rightarrow 2} \frac{z^3 + 2z}{(z^2+1)}$$

$$= \frac{2^3 + 2 \cdot 2}{2(2^2+1)}$$

$$= \frac{12}{5} \neq 0$$

∴ Classification: $f(z)$ has a pole order of 1 at $z=2$

when, $z=i$,

$$\begin{aligned}\lim_{z \rightarrow i} \cancel{(z-i)} \frac{z^3+2z}{(z-2)(z+i)\cancel{(z-i)}} \\&= \lim_{z \rightarrow i} \frac{z^3+2z}{(z-2)(z+i)} \\&= \frac{i^3+2i}{(i-2)(i+i)} \\&= \frac{-i+2i}{-2-4i} \neq 0\end{aligned}$$

$\therefore$ Classification: $f(z)$ has a simple pole
at $z=i$

when, $z=-i$

$$\begin{aligned}\lim_{z \rightarrow -i} \cancel{(z+i)} \frac{z^3+2z}{(z-2)\cancel{(z+i)}(z-i)} \\&= \lim_{z \rightarrow -i} \frac{z^3+2z}{(z-2)(z-i)} \\&= \frac{-i^3+2(-i)}{(-i-2)(-i-i)} \\&= \frac{i-2i}{-2+4i} \neq 0\end{aligned}$$

$\therefore$ Classification: $f(z)$ has a ~~single~~ pole
at $z=-i$ simple

problem 3: $f(z) = \frac{z^L + 1}{(z+1)(z-i)^L}$

when, denominator is ^{every} zero:

$$(z+1)(z-i)^L = 0$$

$\therefore f(z)$ has singular points at $z = -1, +i$

when, $z = -1$

$$\lim_{z \rightarrow -1} \cancel{(z+1)} \frac{z^L + 1}{(z+1)(z-i)^L}$$

$$= \lim_{z \rightarrow -1} \frac{z^L + 1}{(z-i)^L}$$

$$= \frac{(-1)^L + 1}{(-1-i)^L}$$

$$= \frac{2}{1+2i-1} = \frac{1}{i} = -i \neq 0$$

$f(z)$ has a simple pole at $z = -1$

when, $z = i$

$$\lim_{z \rightarrow i} (z-i)^L \frac{z^L + 1}{(z+1)(z-i)^L}$$

$$= \frac{(i)^L + 1}{(i+1)} = 0$$

$$= \lim_{z \rightarrow i} (z-i) \frac{(z+i)(z-i)}{(z+1)(z-i)^L}$$

$$= \lim_{z \rightarrow i} \frac{z+i}{z+1}$$

$$= \frac{i+i}{i+i}$$

$$= \frac{2i}{1+i} \neq 0$$

$\therefore f(z)$ has a simple pole $z = i$

$\therefore f(z)$ is analytic at z -plane except when $z = -1, i$.

problem 5: $f(z) = \frac{(z-1)^3}{z^4 + 4z^2} = \frac{(z-1)^3}{z^2(z+2i)(z-2i)}$

when denominator equals to zero.

$$z^2(z+2i)(z-2i) = 0$$

$f(z)$ has singular points at $z = 0, 2i, -2i$

At, $z = 0$

$$\lim_{z \rightarrow 0} (z-0)^2 \frac{(z-1)^3}{z^2(z+2i)(z-2i)}$$

$$= \lim_{z \rightarrow 0} \frac{(z-1)^3}{(z+2i)(z-2i)}$$

$$= \frac{(0-1)^3}{(0+2i)(0-2i)} = \frac{-1}{4} \neq 0$$

$\therefore f(z)$ has a double pole at $z = 0$

At, $z = 2i$

$$\lim_{z \rightarrow 2i} (z-2i) \frac{(z-1)^3}{z^2(z+2i)(z-2i)}$$

$$= \lim_{z \rightarrow 2i} \frac{(z-1)^3}{z^2(z+2i)}$$

$$= \frac{(2i-1)^3}{-2i(2i)^4(2i+2i)}$$

$$= \frac{(2i-1)^3}{-16i}$$

$$= \frac{-11+2i}{-16i} \neq 0$$

$\therefore f(z)$ has a simple pole at $z = 2i$

At $z = -2i$

$$\lim_{z \rightarrow -2i} (z+2i) \frac{(z-1)^3}{z^4(z+2i)(z-2i)}$$

$$= \lim_{z \rightarrow -2i} \frac{(z-1)^3}{z^4(z-2i)}$$

$$= \frac{(-2i-1)^3}{(-2i)^4(-2i-2i)}$$

$$= \frac{-11-2i}{-16i} \neq 0$$

$\therefore f(z)$ has a simple pole at $z = -2i$

$\therefore f(z)$ is analytic on z -plane except when $z = 0, 2i, -2i$

Residue of a pole

problem: 1 given,

$$f(z) = \frac{3z+1}{(z^2-2)^2(z+1)}$$

The poles are obtained by solving the eq.

$$(z-2)^2(z+1) = 0$$

$$\therefore z = -1, 2$$

Hence, $z = -1$ is simple pole and $z = 2$ is a double pole.

$$R(f, -1) = \lim_{z \rightarrow -1} (z+1) \frac{3z+1}{(z-2)^2(z+1)}$$

$$= \lim_{z \rightarrow -1} \frac{3z+1}{(z-2)^2}$$

$$= \frac{3(-1)+1}{(-1-2)^2} = -\frac{2}{9}$$

$$R(f, 2) = \frac{1}{1!} \lim_{z \rightarrow 2} \frac{d}{dz} \left[\frac{3z+1}{(z-2)^2(z+1)} \right]$$

$$= \lim_{z \rightarrow 2} \frac{d}{dz} \left[\frac{3z+1}{z+1} \right]$$

$$= \lim_{z \rightarrow 2} \frac{3z+3 - 3z-1}{(z+1)^2}$$

$$= \frac{2}{(2+1)^2} = \frac{2}{9}$$

Hence, $R(f, -1) = -\frac{2}{9}$ and $R(f, 2) = \frac{2}{9}$

Q3. given,

$$f(z) = \frac{z^4 + 1}{(z-1)^3}$$

$$(z-1)^3 = 0$$

$$\therefore z = 1$$

$f(z)$ has a simple pole at $z=1$

$$R(f, 1) = \frac{1}{2!} \lim_{z \rightarrow 1} \frac{d^2}{dz^2} \left[(z-1)^3 \frac{z^4 + 1}{(z-1)^3} \right]$$

$$= \frac{1}{2} \lim_{z \rightarrow 1} \frac{d^2}{dz^2} (z^4 + 1)$$

$$= \frac{1}{2} \lim_{z \rightarrow 1} \frac{d}{dz} (4z^3)$$

$$= \frac{1}{2} \lim_{z \rightarrow 1} 12z^2$$

$$= 6$$

Hence, $R(f, 1) = 6$

Q5. given,

$$f(z) = \frac{\sin z}{(z - \frac{\pi}{2})^2}$$

$$z - \frac{\pi}{2} = 0$$

$$\therefore z = \frac{\pi}{2}$$

$f(z)$ has a simple pole at $z = \frac{\pi}{2}$

$$R(f, \frac{\pi}{2}) = \frac{1}{1!} \lim_{z \rightarrow \frac{\pi}{2}} \frac{d}{dz} \left[(z - \frac{\pi}{2}) \frac{\sin z}{(z - \frac{\pi}{2})^2} \right]$$

$$= \lim_{z \rightarrow \frac{\pi}{2}} \cos z$$

$$= \cos\left(\frac{\pi}{2}\right)$$

$$= 0$$

$$\therefore R\left(\frac{\pi}{2}\right) = 0$$

Contour Integral.

① given,

$$\oint_C \frac{dz}{z^2+9}$$

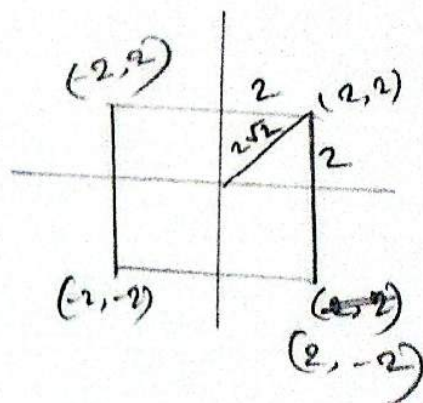
$$z^2+9=0$$

$$\Rightarrow z = \pm 3i$$

$$\Rightarrow \cancel{a = \pm 3i}$$

$$\therefore a = \pm 3i$$

$$|a| = 3$$



C is a closed curve of a square centered at origin and $|a|=3$; But the maximum distance of the closed curve C is $2\sqrt{2}$, which is small than $|a|=3$ outside of the curve. So applying Cauchy's theorem:

$$\oint_C \frac{dz}{z^2+9} = 0$$

(iii) given,

$$\oint_C \frac{dz}{(z^4+1)(z^4+9)} = \oint_C \frac{dz}{(z+i)(z-i)(z^4+9)}$$

$$(z^4+1)(z^4+9) = 0$$

$$\therefore z = \pm i, \pm 3i$$

when $z = \pm 3i$ it is outside of the closed curve as (i).

Residue at $z = i$

$$\begin{aligned} R(f, i) &= \lim_{z \rightarrow i} (z-i) \frac{1}{(z+i)(z-i)(z^4+9)} \\ &= \lim_{z \rightarrow i} \frac{1}{(z+i)(z^4+9)} \\ &= \frac{1}{(i+i)(i^4+9)} \\ &= \frac{1}{16i} = -\frac{i}{16} \end{aligned}$$

$$\begin{aligned} R(f, i) &= \lim_{z \rightarrow -i} (z+i) \frac{1}{(z+i)(z-i)(z^4+9)} \\ &= \lim_{z \rightarrow -i} \frac{1}{(z-i)(z^4+9)} \\ &= \frac{1}{(-i-i)(i^4+9)} = \frac{1}{-16i} = \frac{i}{16} \end{aligned}$$

Applying Cauchy's residue theorem:

$$\begin{aligned}\oint_C \frac{dz}{(z^4+1)(z^4+9)} &= 2\pi i (\text{sum of residues}) \\ &= 2\pi i \left(-\frac{i}{16} + \frac{i}{16}\right) \\ &= 0 \\ &\quad (\text{Ans.})\end{aligned}$$